

ITEC 50: Web Systems and Technologies
Final Project — Group Progress Update Report

Project Title: DeciMark

Members: Lyra Phasma

1 Project Overview

DeciMark is a browser-based Bookmark Link Manager — a client-side web application that enables users to save, organize, retrieve, and annotate URLs using the **Johnny.Decimal** classification system. It requires no server, no account, and no installation, running entirely as a static HTML/CSS/JS artifact in any modern browser. Target users are students and professionals who need structured, consistent link organization without the overhead of cloud-dependent platforms.

2 Current Progress

Conceptualization and competitive analysis are complete. Core design philosophy and feature differentiation from existing tools — including Pinboard, linkding, and Raindrop.io — have been fully documented. The Johnny.Decimal classification scheme has been selected and justified as the primary organizational model, and the project's scope has been deliberately bounded to a zero-dependency, offline-first static implementation.

3 Planned Tasks

- Implementation of the card-based UI with list-view toggle
- Johnny.Decimal ID field integration and range-based filtering
- Tag support and multi-attribute search functionality
- JSON-based local data persistence
- Final polish, accessibility improvements, and export functionality

4 Tools and Technologies

- **Languages:** HTML, CSS, JavaScript, Python
- **Design reference:** Johnny.Decimal organizational system (<https://johnnydecimal.com>)
- **No external frameworks or backend dependencies** — by design

5 Challenges / Concerns

Working as a sole member required managing all responsibilities — research, design, documentation, and implementation — independently. This was not without its silver linings, however. A concurrent hackathon project provided substantial hands-on experience in full-stack development, database architecture, backend frameworks, and deployment pipelines, all of which have meaningfully deepened my understanding of web systems at a level that directly informs the execution of DeciMark.

6 Additional Notes

It was not all in vain.

The hackathon experience — undertaken in parallel with this course — resulted in the development of **Pebble Recall**, a full-stack psychiatric care companion application built with Python, FastAPI, SQLAlchemy, PostgreSQL, and Jinja2, deployed live at <https://pebble-recall.lyra-on.top>.

Pebble Recall placed **#1 in judge's choice** and **#1 in community vote** out of more than 140 competing teams, winning the **₱20,000 grand prize**.

The skills developed over the course of that project — including database migration with Alembic, asynchronous backend architecture, authentication systems, and full deployment pipelines — will directly strengthen the technical execution of DeciMark going forward. The detour, it turns out, was not a detour at all.